use would be approved quickly enough to stop frog infestations. As such, one aspect of the current invention is to improve the frog killing effects of compounds that are already listed as EPA approved pesticide ingredients, or for frog eradication.

More specifically, compounds such as propionic acid, which are already approved as pesticides, can be used to take advantage of methods described herein to titrate lower pH levels for the purpose of killing frogs, without being as toxic to plants. Propionic acid (systematically named propanoic acid) is a naturally-occurring carboxylic acid with chemical formula CH_3CH_2COOH. It is a clear liquid with a pungent odor. The anion CH_3CH_2COO as well as the salts and esters of propionic acid are known as propionates (or propanoates) and are considered to be functional chemical equivalents thereof.

Although not wanting to be bound by current regulations and/or theory, ACS should be formulated to meet the HAZMAT cutoff of pH 2.0, however, titrating ACS with propionic acid or propionic acid salts can be utilized to increase or decrease the pH. In contrast, properly formulated propionic acid is at least as effective as ACS alone in killing frogs, and can be formulated will little effort to have a pH within the HAZMAT regulations of about 2.0. Unfortunately, an increase in concentration may lead to increased phytotoxicity. More specifically, propionic acid is available as a dry product and could be used in much the same way as citric acid is now being used and would cost less per gallon when compared to citric acid. For example, 1 gallon of 16% citric acid costs about $2.40/gallon (retail) and is considered phytotoxic to many plants, whereas, 1 gallon of 16% propionic acid can be formulated to have a lower pH and costs only about $1.00 per gallon with similar or superior phytotoxicity.

Although EPA regulations can present hurdles for entry of new compositions into the pesticide market, citric acid and/or propionic acid are currently products that are on EPAs list 4 and can safely be used in pesticides without concern for non-target species and or negative impact on the water table. As such, modified formulas of citric acid having lower phtotoxicity may also have a lower hurdle for EPA approval. More specifically, it is possible to use ACS in a synergistic way with citric acid to actually lower the 16% concentration that is know to have phytotoxic properties, while retaining the capacity to kill the frogs. For this reason, it may be advantageous to lower the 16% citric acid concentration in order to lower its known phytotoxic effects.

Similarly, a composition of propionic acid is prepared as a liquid using any of the following three methods shown below. The active ingredient is proprionic acid and either ACS or sulfurinc acid is used to adjust the pH downward or one of the propionic salts to adjust the pH upward. ACS-P consists of calcium sulfate saturated with sulfuric acid, can also be used to adjust the pH and/or to make slurry. ACS-P can be dried and/or infused into bentonite clay, vermiculite, or some other functional equivalent carrier capable of making a dry product. All formulations can be made to vary in concentration according to the needs and applications of the user or EPA standards. For example, the pH of the final product solution will for most purposes be at 2.0 or above in order to meet the HAZMAT criteria for non-hazardous products.

Formulation 1. Propionic acid in water.

To make an approximate 10% solution, add about 5-15 ml of propionic acid (90%+purity) to 100 ml of water. Mix and check pH to make sure it falls within the range from 2.0 to 2.5. If pH is greater than 2.5, titrate pH downward by adding a small amount (0.25 to 1.0 ml) of sulfuric acid (90%+purity). If pH is less than 2.0 adjust pH upward using a small amount (+/−about 0.5 to 1.0 gram) of sodium propionate. In a one embodiment, the concentration range of propionic acid is in the range of about 5.0% to about 15%, and the pH is in the range of about 1.5 to about 3.5. In a preferred embodiment, the concentration range of propionic acid is in the range of about 8.0% to about 12.5%, and the pH is in the range of about 1.8 to 2.8. In the most preferred embodiment, the concentration range of propionic acid is in the range of about 9.5% to about 10.5%, and the pH is in the range of about 2.2 to about 2.5. Alternatively, the pH can be titrated for the various concentration ranges using a superacid such as ACS-100, ACS-50, sulfuric acid, hydrochloric acid, or combination thereof (Note: a pipette or syringe can be used to slowly add sulfuric acid for adjusting pH downward to between about pH 2.0 and about pH 2.5. In some instances, it may be unadvisable to allow the total percentage of sulfuric acid used to adjust pH be above about 10% of the total solution, as not to surpass EPA tolerance limit for the use of sulfuric acid as a pH adjuster in pesticides. Additionally, propionic acid is a volatile chemical and may evaporate out of solution within a given period of time. It is therefore advisable to use methods known in the art to prevent such evaporation from occurring. One having ordinary skill in the art will understand that methods and ratios used to industrialize large scale manufacturing protocols of these compositions is included in the spirit and scope of the invention.

Formulation 2. Salt of Propionic Acid in Water.

To make an approximate 10% solution, add about 10 grams of either sodium or calcium propionate to about 100 ml of water, wherein sodium propionate is preferred because it does not leave a precipitant. The initial pH should be in the range of about 4.0 to 6.0. Adjust pH range to about 2.0-about 2.5 using sulfuric acid, ACS-P, ACS-100, or ACS-50. In a one embodiment, the concentration range of the salt of propionic acid is in the range of about 5.0% to about 15%, and the pH is in the range of about 1.5 to about 3.5. In a preferred embodiment, the concentration range of the salt of propionic acid is in the range of about 8.0% to about 12.5%, and the pH is in the range of about 1.8 to about 2.8. In the most preferred embodiment, the concentration range of the salt of propionic acid is in the range of about 9.5% to about 10.5%, and the pH is in the range of about 2.2 to about 2.5. Alternatively, the pH can be titrated for the various concentration ranges using a superacid, ACS-100, ACS-50, sulfuric acid, or hydrochloric acid. One having ordinary skill in the art will understand that methods and ratios used to industrialize large scale manufacturing protocols of these compositions is included in the spirit and scope of the invention.

Formulation 3. Acs-P and Acs-100 in Water.

ACS-P and ACS 100 in water with or without the use of any other ingredients to adjust the pH of the solution, in one embodiment of this invention, an approximate 10% solution can be made by adding about 10 grams of ACS-P 50 to 100 ml of tap water and mix thoroughly. The pH should be should be in the range of about pH 1.1 to about pH 1.3. The pH can be adjusted within the desired range of about pH 1.0 by adding a few ml of ACS. If the pH adjustment results in a pH that is below 1.0, a small amount of calcium hydroxide can be added to increase the pH. The clear liquid can be decanted from any precipitant; the decanted liquid can be used as a spray. Save the white precipitant created in this process for use as a slurry product as described in Formulation 4 described below.

Since, the pH of this solution is about pH 1.0, it may harm certain plants by turning the leaves yellow, wherein most plants should recover and plants should not die from application of this product. Although not wanting to be bound by theory, this product should be maintained in a range between